MILITARY SPECIFICATION SHEET

ELECTRON TUBE, KLYSTRON

TYPE 6116

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: CW, reflex oscillator, integral cavity, thermally tuned, frequency range 8,500 to 9,660 MHz

ABSOLUTE RATINGS:

Parameter:	Ef	Ers	Er	Pp	Ik1	Ebk1	Eb	Alt
Unit:	V	Vdc	Vdc	W	mAdc	Vdc	Vdc	ft
Maximum:	6.8	350	-350	16.5	32	±100	350	10,000
Minimum:	5.8 (see note 5)	---	---	--- (see notes 1 and 5)	--- (see note 5)	--- (see note 5)	--- (see note 5)	---

PHYSICAL CHARACTERISTICS:

Dimensions:	See figure 1
Cathode:	Unipotential
Mounting:	See note 2

TEST CONDITIONS:

Ef1	Ef2	Ers	Er	Eb
V	V	Vdc	Vdc	Vdc
6.3	6.3	300	-60 to -175	0 to 280

Frequency		
F	MHz	±
F1	8,500	0.3%
F2	9,080	0.3%
F3	9,660	0.3%

GENERAL:

Qualification - Required

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 1	6					
1256	Electrode current (cathode 1)	5	Er = -150 Vdc	Ik1	---	32	mAdc
4250	Power output	5, 7	Ef1 = 5.8 V; F = F1 to F3	Po	20	---	mW
4213	Reflector voltage	7	F = F3	Er	-95	-145	Vdc
4214	Cathode emission	---					
	Cathode 1	5	Ef1 = 6.3 to 5.8 V; t = 120 sec; Er = -150 Vdc	Δik1/Ik1	---	15	%
	Cathode 2	5	Ef2 = 6.3 to 5.8 V; t = 120 sec; Ik2 = 20 mAdc	Δik2/Ik2	---	10	%
4229	Total reflector current	---	Er = -150 Vdc; t = 120 sec (min)	Ir	---	5.0	μAdc
4229	Reflector-leakage current	3	Er = -150 Vdc	Ir	---	5.0	μAdc
4229	Reflector-gas current	---	Er = -150 Vdc	Ir	---	2.0	μAdc
---	Thermal tuning time	---					
	Tuning time 1	4	F = F3 to F1	t	0.7	3.0	sec
	Tuning time 2	4	F = F1 to F3	t	0.7	3.0	sec
	Quality conformance inspection, part 2						
1301	Heater current 1	5		If1	465	570	mA
1301	Heater current 2	5		If2	720	880	mA
1211	Insulation of electrodes	---	Ers = 300 Vdc; tube cold	Rk1-rs Rk2-rs Rh1-rs	2.0 2.0 2.0	--- --- ---	Meg Meg Meg
1336	Heater-cathode leakage	---	Ehk1 = +100 Vdc	Ihk1	---	100	μAdc
1031	High-frequency vibration	7	F = F2	ΔF	---	±1,300	kHz
---	Electronic tuning range	---	Er (mode A)/50% max Po; F = F1 to F3	ΔF	45	---	MHz
---	Thermal tuning range	5, 7	Ef1 = 5.8 V; Pp = 1.0 to 9.0 W				
			High frequency	F	9,660	---	MHz
			Low frequency	F	---	8,500	MHz
---	Tuner diode voltage drop	---	Ik2 = 10 mAdc	Eb	170	218	Vdc
			Ik2 = 28 mAdc	Eb	225	274	Vdc
---	Tuner diode voltage	7	F = F3	Eb	170	230	Vdc
			F = F2	Eb	200	260	Vdc
			F = F1	Eb	220	275	Vdc

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 2 -Continued						
---	Tuner diode current	7	F = F3	Ik2	5	19	mAdc
			F = F2	Ik2	13	28	mAdc
			F = F1	Ik2	20	36	mAdc
	Quality conformance inspection, part 3						
---	Life-test provisions	7	Group C; Ik2 = 20 mAdc	t	500	---	hrs
---	Life-test end points:	---					
4250	Power output	5, 7	Ef1 = 5.8 V; F = F1 to F3	Po	10	---	mW
---	Thermal tuning range	7	Ef1 5.8 V, Pp = 0.5 to 9.5 W				
			High frequency	F	9,660	---	MHz
			Low frequency	F	---	8,500	MHz

NOTES:

1. Power inputs as high as 16.5 watts may be applied to the diode when the frequency of the klystron is above 8,500 MHz. Tuner power in excess of 10 watts may permanently damage the tuning structure if applied when the tube is tuned below 8,500 MHz.

2. The tube shall be fixed firmly in a suitable socket by clamps which make contact with the tube at the base and adjacent platform only. The measurements on the tube in an oscillating state shall be made with the output lead coupled into a waveguide tube mount section per Drawing 227-1AN, or equivalent. Measurements taken with the tube oscillating shall be done with a matched load. (VSWR = 1.15:1 maximum.)

3. The cathode lead to be opened shall be cathode 1 in accordance with method 4229.

4. The thermal tuning speed shall be measured in the following manner: The tuner diode voltage and reflector voltage shall be mutually adjusted to produce maximum power under stable conditions of the initial frequency indicated for thermal tuning time 1 or thermal tuning time 2. The tuner diode voltage and reflector voltage shall then be switched simultaneously to the end condition values indicated in the table. The time interval between the instant of switching and the time at which the tube delivers maximum output at the end frequency of the test shall not exceed the specified value.

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NOTES: -Continued

Test	Condition	Eb and Er conditions	F
Thermal tuning time 1	Initial	Mutually adjusted for max Po in Mode A at initial frequency	9,660 MHz ±0.3%
	End	1/ Eb = 280 Vdc; Er(Mode A)/max Po at 8,500 MHz	8,500 MHz ±0.3%
1/ The diode plate potential Eb shall not be held at 280 volts for longer than 2 seconds after the end frequency (8,500 MHz) has been reached.			
Thermal tuning time 2	Initial	Mutually adjusted for max Po in Mode A at initial frequency	8,500 MHz ±0.3%
	End	Eb = 0; Er(Mode A)/max Po at 9,660 MHz	9,660 MHz ±0.3%

5. Cathode 1, Ef1, and heater current 1 refer to parts associated with the radio frequency section of the tube. Cathode 2, Ef2, and heater current 2 refer to parts associated with the thermal tuner diode section of the tube. Symbol Pp refers to power dissipation in the thermal tuner anode. Symbol Eb refers to the plate potential of the thermal tuner diode.

6. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0 percent, with inspection level II.

7. Er (mode A) max Po.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - MU
Air Force - 11, 17
DSA - ES

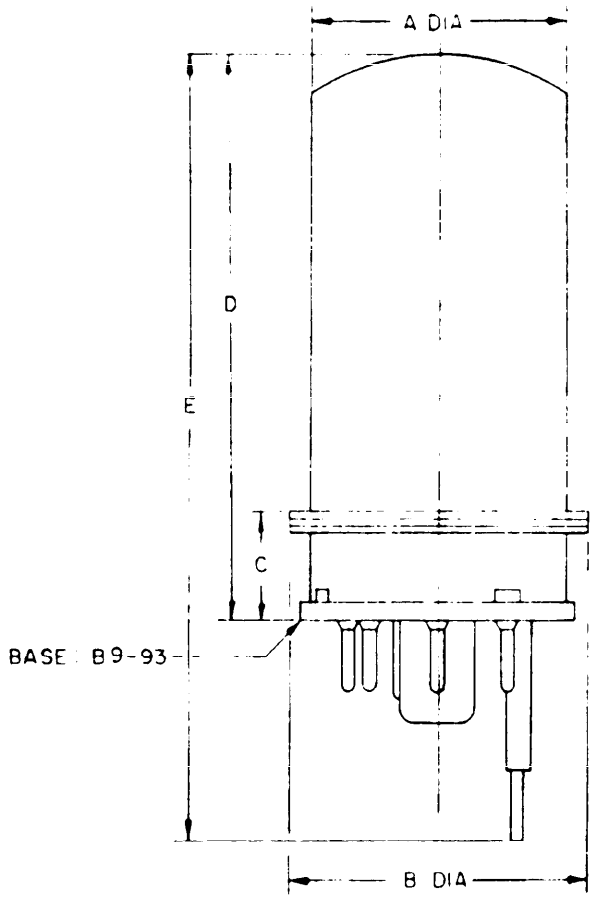
User activities:
Army - WC
Navy - AS, OS, CG, SH

Preparing activity: Navy - EC

Agent: DSA - ES

(Project 5960-2425-38)

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, Part 1 (See note 6)		
A	———	1.281 (32.54)
B	1.500 (38.10)	1.625 (41.28)
C	.375 (9.53)	.437 (11.10)
D	———	2.562 (65.07)
E	———	3.796 (96.42)

BASE CONNECTIONS	
PIN NO.	ELEMENT
1	Tuner heater
2	Tuner heater-k
3	rs and sh
4	Output
5	Gun-k
6	Heater (1)
7	Heater (1)
8	Reflector

FIGURE 1. Outline drawing of electron tube type 6116.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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